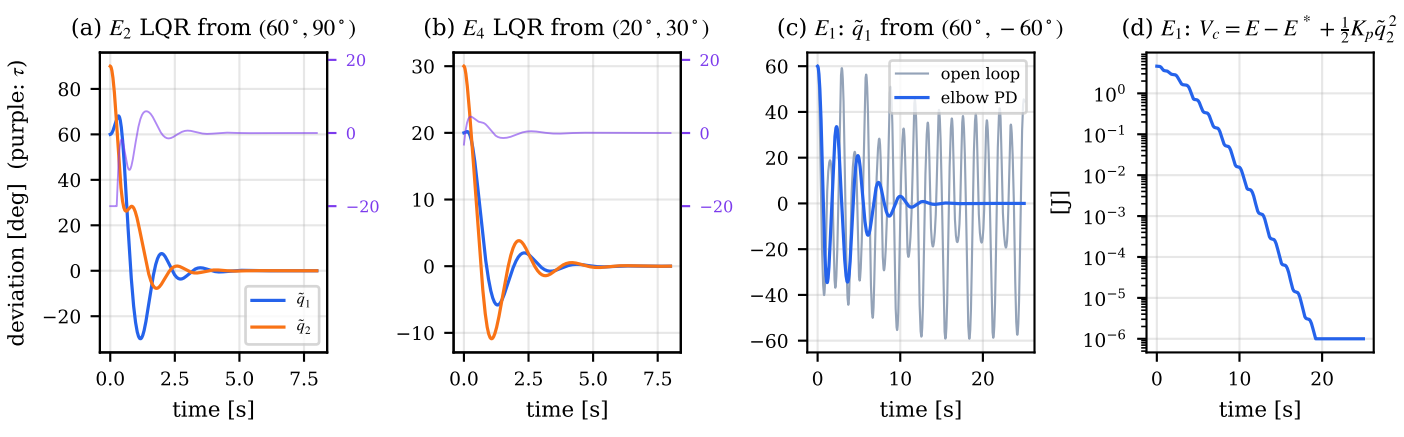
(a) E_2 LQR from $(60^\circ, 90^\circ)$
(b) E_4 LQR from $(20^\circ, 30^\circ)$
(c) E_1: $\tilde{q}_1$ from $(60^\circ, -60^\circ)$
(d) E_1: $V_c = E - E^* + \frac{1}{2}K_p\tilde{q}_2^2$
deviation [deg] (purple: τ)
time [s]
$\tilde{q}_1$
$\tilde{q}_2$
open loop
elbow PD
[J]